3.4.7 Ecosystems are dynamic systems, usually moving from colonisation to climax communities in the process of succession.

Succession	Succession from pioneer species to climax community. At each stage in succession, certain species may be recognised which change the environment so that it becomes more suitable for other species. The changes in the abiotic environment result in a less hostile environment and changing diversity. Conservation of habitats frequently involves management of succession. **Candidates should be able to** • use their knowledge and understanding to present scientific arguments and ideas relating to the conservation of species and habitats • evaluate evidence and data concerning issues relating to the conservation of species and habitats and consider conflicting evidence • explain how conservation relies on science to inform decision-making.

3.4.8 Genetic variation within a species and geographic isolation leads to the accumulation of different genetic information in populations and the potential formation of new species.

Inheritance	The genotype is the genetic constitution of an organism. The phenotype is the expression of this genetic constitution and its interaction with the environment. The alleles at a specific locus may be either homozygous or heterozygous. Alleles may be dominant, recessive or codominant. There may be multiple alleles of a single gene. **Candidates should be able to** use fully labelled genetic diagrams to predict the results of • monohybrid crosses involving dominant, recessive and codominant alleles • crosses involving multiple alleles and sex-linked characteristics.
The Hardy-Weinberg principle	Species exist as one or more populations. The concepts of gene pool and allele frequency. The Hardy-Weinberg principle. The conditions under which the principle applies. **Candidates should be able to** calculate allele, genotype and phenotype frequencies from appropriate data and from the Hardy-Weinberg equation, $p^2 + 2pq + q^2 = 1$ where p is the frequency of the dominant allele and q is the frequency of the recessive allele. **Candidates should understand that** the Hardy-Weinberg principle provides a mathematical model which predicts that allele frequencies will not change from generation to generation.

Selection	Differential reproductive success and its effect on the allele frequency within a gene pool. Directional and stabilising selection. **Candidates should be able to** • use both specific examples and unfamiliar information to explain how selection produces changes within a species • interpret data relating to the effect of selection in producing change within populations.
Speciation	Geographic separation of populations of a species can result in the accumulation of difference in the gene pools. The importance of geographic isolation in the formation of new species

Biological principles

When they have completed this unit, candidates will be expected to have an understanding of the following principles.

- Random sampling results in the collection of data which is unbiased and suitable for statistical analysis
- ATP is the immediate source of energy for biological processes
- Limiting factors
- Energy is transferred through ecosystems
- Chemical elements are recycled in ecosystems
- Genetic variation occurs within a species and geographic isolation leads to the accumulation of genetic difference in populations.

Understanding of these principles may be tested in Units 5 and 6. Such testing may contribute to the assessment of synoptic skills.

Investigative and practical skills

Opportunities to carry out the necessary practical work should be provided in the context of this unit. Candidates will be assessed on their understanding of *How Science Works* in all three A2 Units. To meet this requirement candidates are required to

- use their knowledge and understanding to pose scientific questions and define scientific problems
- carry out investigative activities, including appropriate risk management, in a range of contexts
- analyse and interpret data they have collected to provide evidence
- evaluate their methodology, evidence and data, resolving conflicting evidence.

The practical work undertaken should include investigations which consider

- the effect of a specific limiting factor such as light intensity, carbon dioxide concentration or temperature on the rate of photosynthesis
- the effect of a specific variable such as substrate or temperature on the rate of respiration of a suitable organism
- fieldwork involving the use of frame quadrats and line transects, and the measurement of a specific abiotic factor; collection of quantitative data from at least one habitat, and the application of elementary statistical analysis to the results; the use of percentage cover and frequency as measures of abundance.